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Inventor(s)	Meckes; Nathan

Snowboard Assistance Device

Abstract

A training device for snowboard users. The training device intends to minimize the chances of a child falling while learning proper snowboarding techniques. The training device comprises a base plate, a handle receiver, a handle, and a locking system. The front of the base plate mounts to the rear bindings of the trainee's snowboard. The handle receiver is integrated into the rear of the base plate. The handle detachably connects to the handle receiver and the locking system secures the handle in place. During use, the trainer rides behind the trainee, wherein the trainer has control of the handle. While in motion, the trainer can twist, push, and pull on the handle to control the speed, edge angle, and rotation of the snowboard. When not in use, the handle detaches from the device. This allows the user to ride on a chairlift with the training device in a stowed configuration.

Inventors: **Meckes; Nathan (Crested Butte, CO)**

Applicant: **Meckes; Nathan** (Crested Butte, CO)

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to snowboard attachments. More specifically, the present invention is a training device that allows an instructor or parent to control the speed, edge angle and rotation of a snowboard.

BACKGROUND OF THE INVENTION

[0002] A snowboard assistance trainer device is a specialized tool designed to help snowboarders, especially beginners, improve their skills and confidence on the slopes. These devices are often used in snowboard schools, training facilities, and by individual riders looking to enhance their abilities. Typically, a snowboard assistance trainer device consists of a sturdy frame or platform with bindings that securely hold the snowboard in place. The device may also include adjustable straps or restraints to ensure the rider's safety and stability while practicing.

[0003] One common type of snowboard assistance trainer device is a balance bar or rail, which extends horizontally from the frame and provides riders with a stable surface to hold onto while practicing balance and maneuvering techniques. This allows beginners to get a feel for the snowboard's movements without the fear of falling. Some snowboard assistance trainer devices also incorporate features such as adjustable inclines or ramps, allowing riders to practice riding downhill and performing turns in a controlled environment. This can help build confidence and muscle memory before tackling more challenging terrain on the mountain. However, these devices do not allow for a user to control another individual's snowboard position and speed. The present invention provides a tool for snowboarders of all skill levels, providing a safe and supportive environment for learning and practicing essential techniques. By offering stability, guidance, and feedback, these devices can help riders improve their skills and progress more quickly on the slopes.

[0004] An objective of the present invention is to provide users with a device that secures to a child's snowboard, to help control the attached snowboard's speed, edge angle and rotation. The present invention intends to minimize the chance of the child falling while learning proper snowboarding technique. To accomplish this objective, a preferred embodiment of the present invention comprises a base plate, a handle receiver, a handle, and a locking system. Further, the handle allows for a user to hold and secure the present invention while attached to a snowboard. The locking system helps secure the handle to the handle receiver during use. Thus, the present invention is a device that secures to the rear bindings of an external snowboard and allows a user to control the speed, edge angle and position of the attached external snowboard.

SUMMARY OF THE INVENTION

[0005] The present invention is a training device that allows the user (trailing rider) to easily push or tilt the attached snowboard. The present invention seeks to provide users with a device that allows a user to teach a child proper snowboarding technique in a safe manner. To accomplish this objective, the present invention comprises a base plate that secures to the rear bindings of a snowboard. A handle receiver mounts to the base plate and receives the handle. Additionally, the handle is detachable for easy transportation and allows the user to easily control the device. A ratchet strap secures the handle to the handle receiver. Thus, the present invention is a device that secures to the rear bindings of an external snowboard and allows the user to control the speed, edge angle and position of the attached snowboard.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a top-front-right perspective view of the present invention, shown attached to a snowboard.

[0007] FIG. **2** is a top-rear-left perspective view of the present invention, shown attached to a snowboard.

[0008] FIG. **3** is a top-front-right perspective view of the present invention.

[0009] FIG. **4** is a bottom-rear-left perspective view of the present invention.

[0010] FIG. **5** is a front elevational view of the present invention.

[0011] FIG. **6** is a rear elevational view of the present invention.

[0012] FIG. **7** is a right-side elevational view of the present invention.

[0013] FIG. **8** is a top plan view of the present invention.

[0014] FIG. **9** is a bottom plan view of the present invention.

[0015] FIG. **10** is a magnified view taken from FIG. **7**, in accordance with a first embodiment.

[0016] FIG. **11** is a magnified view taken from FIG. **7**, in accordance with a second embodiment.

DETAIL DESCRIPTIONS OF THE INVENTION

[0017] All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.

[0018] In reference to FIGS. **1-11**, the present invention is a training device **1** for snowboard users. An objective of the present invention is to provide the user with a training device that is easily transportable and detachable. The training device **1** mounts to the rear bindings **9** of a snowboard SB. While in motion, the user (trailing rider) twists, pushes, and pulls on the handle **4** of the training device to control the speed, edge angle, and rotation of the snowboard. When not in use, the handle **4** detaches from the device. This allows the user to ride on a chairlift with the training device **1** in a stowed configuration.

[0019] To accomplish the objectives, the present invention comprises a base plate **2**, a handle receiver **3**, a handle **4**, and a locking system **5**. The base plate **2** functions as the primary structural component of the present invention, as the remaining components of the present invention are configured upon the base plate **2**. The base plate **2** is a rigid structure made of metal or hard plastic and is preferably rectangular in shape. As best seen in FIGS. **8-10**, the base plate **2** comprises a top surface **21**, a bottom surface **22**, a forward section **23**, and an aft section **24**. The bottom surface **22** of the base plate is seated flush with the top of the snowboard SB. The forward section **23** of the base plate is secured to the rear bindings **9** of the snowboard SB. The handle receiver **3** is positioned within the aft section **24** of the base plate, extending outward from the top surface **21** at a predefined angle α. The handle **4** is detachably connected to the handle receiver **3**, and the locking system **5** is configured to secure the handle **4** to the handle receiver **3**. When preparing for transport, the user can quickly detach the handle **4** from the handle receiver **3** by disengaging the locking system **5**.

[0020] In a preferred embodiment, the same OEM fasteners used to connect the rear bindings **9** to the snowboard SB are also utilized to detachably connect the base plate **2** to the snowboard. In this embodiment, as seen in FIG. **3**, the base plate **2** further comprises a plurality of binding holes **25**. The plurality of binding holes **25** are disposed along the forward section **23** of the base plate **2**. The forward section **23** is positioned in between the top of the snowboard SB and the bottom of the rear bindings **9**. A plurality of first fasteners **91** are inserted through the rear bindings **9**, through the corresponding binding holes **25** of the base plate, and fastened into the snowboard SB. The exact quantity and positioning of the binding holes **25** on the base plate are delineated by the mounting provisions of the rear bindings **9**.

[0021] The handle receiver **3** comprises a receiver body **31** and a receiver opening **32**. As seen in FIG. **10**, the receiver body **31** extends upward and away from the base plate **2** at an obtuse angle α, wherein the obtuse angle α is delineated between the top surface **21** of the base plate **2** and the front of the receiver body **31**. The receiver opening **32** traverses axially inward from the top of the receiver body **31**. The receiver opening **32** is preferably in the form of a hexagonal-shaped counterbore. The hexagonal shape prevents the handle **4** from rotating during use. It is understood that the angle α of the receiver body is not limited to being an obtuse angle and can be of any suitable angle based on design, user, and/or manufacturing requirements. Likewise, the profile shape of the receiver opening **32** is not limited to hexagonal and can take the form of any suitable shape based on design, user, and or manufacturing requirements.

[0022] In the preferred embodiment, the handle receiver **3** further comprises a plurality of brace supports **33**. The plurality of brace supports **33** provide structural stability to help strengthen the connection between the base plate **2** and the handle receiver **3**. During use, the plurality of brace supports **33** absorb the high stress and high torque loads exerted from the handle **4**. Each of the plurality of brace supports **33** are perimetrically connected along the outer surface of the receiver body **31**, extending downward to the top surface **21** of the base plate **2**. The plurality of brace supports **33** comprises a first side brace **34**, a second side brace **35**, and an aft brace **36**. As best seen in FIGS. **5-6**, the first and second side braces **34**,**35** are positioned on opposite sides of the handle receiver **3**, each oriented perpendicular with the aft brace **36**. The aft brace **36** is positioned along the aft end **26** of the base plate **2**. In this arrangement, the aft brace **36** provides vertical support while the first and second side braces **34**,**35** provide lateral support.

[0023] The handle **4** comprises a handle stem **41** and a handle grip **42**. The handle stem **41** is an elongated member, preferably in the form of a cylindrical tube. However, the shape of the handle stem **41** is not limited to being cylindrical and can take the form of any suitable shape based on design, user, and/or manufacturing requirements. The handle stem **41** comprises a connector end **40** and a distal end **49**. As seen in FIG. **10**, the connector end **40** is adapted to slidably engage with the receiver opening **32**. Stated another way, the connector end **40** is shaped to fit within the receiver opening **32**. The distal end **49** of the handle stem **41** is terminally connected to the handle grip **42**. In the preferred embodiment, as seen in FIG. **8**, the handle grip **42** is configured as a two-handed grip comprising a pair of side handles **42***a*, **42***b* and a center handle **42***c*. In this arrangement, the center handle **42***c* allows the user to push the snowboard forward or pull the snowboard back with one hand. The pair of side handles **42***a*, **42***b* allow the user to provide torque and tilt the snowboard in either direction using two hands.

[0024] In the preferred embodiment, the locking system **5** is in the form of a ratchet strap **5***a*. As seen in FIG. **10**, the ratchet strap **5***a* is easily accessible, allowing the user to quickly engage and disengage the ratchet strap **5***a* without using any external tools. The ratchet strap **5***a* further comprises a ratchet **51**, a rubber strap **52**, and a second fastener **53**. The ratchet **51** is mounted along the outer surface of the handle stem **41**, positioned above the connector end **40**. A first end **52***a* of the rubber strap **52** is mounted to the handle receiver **3** via the second fastener **53**. The second end **52***b* of the rubber strap **52** is inserted through and operably coupled to the ratchet **51**. To unlock and remove the handle **4**, the user simply pulls up on the lever of the ratchet **51** and slides the handle **4** out of the handle receiver **3**.

[0025] In an alternative embodiment, the locking system **5** is in the form of a lock bolt **5***b*. As seen in FIG. **11**, a plurality of apertures **54** traverse laterally through the receiver body **31** and through the connector end **40**. To secure the handle **4** in place, the user inserts the connector end **40** into the receiver opening **32** such that the apertures **54** on the connector end **40** are aligned with the apertures **54** on the receiver body **31** to form a through hole. Then, the user inserts the lock bolt **5***b* through the plurality of apertures **54**, thereby securing the handle **4** in place. To unlock and remove the handle **4**, the user follows the same steps in reverse.

[0026] Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention.

Claims

1. A snowboard training device comprising: a base plate; a handle receiver; a handle; a locking system; the base plate being positioned on top of a snowboard; the base plate comprising a forward section and an aft section; the forward section being secured to the rear bindings of the snowboard; the handle receiver being positioned within the aft section; the handle receiver extending outward from the top surface of the base plate at a predefined angle; the predefined angle being delineated between the top surface of the base plate and the front of the handle receiver; the handle being detachably connected to the handle receiver; and the locking system being configured to secure the handle to the handle receiver.

2. The snowboard training device as claimed in claim 1 comprising: the base plate further comprising a plurality of binding holes; the plurality of binding holes being positioned along the forward section; the forward section being positioned in between the top of the snowboard and the rear bindings; and a plurality of first fasteners being inserted through the rear bindings, through corresponding binding holes of the base plate, and fastened into the snowboard.

3. The snowboard training device as claimed in claim 1 comprising: the handle receiver comprising a receiver body and a receiver opening; and the receiver opening traversing axially inward from the top of the receiver body.

4. The snowboard training device as claimed in claim 3, wherein the predefined angle is an obtuse angle.

5. The snowboard training device as claimed in claim 3 comprising: the handle receiver further comprising a plurality of brace supports; each of the plurality of brace supports being perimetrically connected to the receiver body, extending downward to the top surface of the base plate; the plurality of brace supports comprising a first side brace, a second side brace, and an aft brace; and the aft brace being positioned along the aft end of the base plate.

6. The snowboard training device as claimed in claim 3 comprising: the handle comprising a handle stem and a handle grip; the handle stem comprising a connector end and a distal end; the connector end being adapted to slidably engage with the receiver opening; and the distal end being terminally connected to the handle grip.

7. The snowboard training device as claimed in claim 6 comprising: the handle grip comprising a pair of side handles and a center handle; the pair of side handles being configured for tilting the snowboard; and the center handle being configured for pushing and pulling the snowboard.

8. The snowboard training device as claimed in claim 6 comprising: the locking system being in the form of a ratchet strap; the ratchet strap comprising a ratchet, a strap, and a second fastener; the ratchet being mounted to the handle stem, positioned above the connector end; a first end of the strap being mounted to the handle receiver via the second fastener; and a second end of the strap being operably coupled to the ratchet.

9. The snowboard training device as claimed in claim 6 comprising: the locking system being in the form of a lock bolt; a plurality of apertures traversing laterally through the receiver body and through the connector end; each of the plurality of apertures being aligned to form a through hole; and the lock bolt being inserted into the plurality of apertures.

10. A snowboard training device comprising: a base plate; a handle receiver; a handle; a locking system; the base plate being positioned on top of a snowboard; the base plate comprising a forward section, an aft section, and a plurality of binding holes; the forward section being secured to the rear bindings of the snowboard; the plurality of binding holes being positioned along the forward section; the forward section being positioned in between the top of the snowboard and the rear bindings; a plurality of first fasteners being inserted through the rear bindings, through corresponding binding holes of the base plate, and fastened into the snowboard; the handle receiver being positioned within the aft section; the handle receiver extending outward from the top surface of the base plate at an obtuse angle; the obtuse angle being delineated between the top surface of the base plate and the front of the handle receiver; the handle receiver comprising a receiver body and a receiver opening; the receiver opening traversing axially inward from the top of the receiver body; the handle being detachably connected to the handle receiver; and the locking system being configured to secure the handle to the handle receiver.

11. The snowboard training device as claimed in claim 10 comprising: the handle receiver further comprising a plurality of brace supports; each of the plurality of brace supports being perimetrically connected to the receiver body, extending downward to the top surface of the base plate; the plurality of brace supports comprising a first side brace, a second side brace, and an aft brace; and the aft brace being positioned along the aft end of the base plate.

12. The snowboard training device as claimed in claim 10 comprising: the handle comprising a handle stem and a handle grip; the handle stem comprising a connector end and a distal end; the connector end being adapted to slidably engage with the receiver opening; and the distal end being terminally connected to the handle grip.

13. The snowboard training device as claimed in claim 12 comprising: the handle grip comprising a pair of side handles and a center handle; the pair of side handles being configured for tilting the snowboard; and the center handle being configured for pushing and pulling the snowboard.

14. The snowboard training device as claimed in claim 12 comprising: the locking system being in the form of a ratchet strap; the ratchet strap comprising a ratchet, a strap, and a second fastener; the ratchet being mounted to the handle stem, positioned above the connector end; a first end of the strap being mounted to the handle receiver via the second fastener; and a second end of the strap being operably coupled to the ratchet.

15. The snowboard training device as claimed in claim 12 comprising: the locking system being in the form of a lock bolt; a plurality of apertures traversing laterally through the receiver body and through the connector end; each of the plurality of apertures being aligned to form a through hole; and the lock bolt being inserted into the plurality of apertures.

16. A snowboard training device comprising: a base plate; a handle receiver; a handle; a locking system; the base plate being positioned on top of a snowboard; the base plate comprising a forward section and an aft section; the forward section being secured to the rear bindings of the snowboard; the handle receiver being positioned within the aft section; the handle receiver extending outward from the top surface of the base plate at a predefined angle; the predefined angle being delineated between the top surface of the base plate and the front of the handle receiver; the handle receiver comprising a receiver body and a receiver opening; the receiver opening traversing axially inward from the top of the receiver body; the handle comprising a handle stem and a handle grip; the handle stem comprising a connector end and a distal end; the connector end being adapted to slidably engage with the receiver opening; the distal end being terminally connected to the handle grip; the handle being detachably connected to the handle receiver; the locking system being configured to secure the handle to the handle receiver; the locking system being in the form of a ratchet strap; the ratchet strap comprising a ratchet, a strap, and a second fastener; the ratchet being mounted to the handle stem, positioned above the connector end; a first end of the strap being mounted to the handle receiver via the second fastener; and a second end of the strap being operably coupled to the ratchet.

17. The snowboard training device as claimed in claim 16 comprising: the base plate further comprising a plurality of binding holes; the plurality of binding holes being positioned along the forward section; the forward section being positioned in between the top of the snowboard and the rear bindings; and a plurality of first fasteners being inserted through the rear bindings, through corresponding binding holes of the base plate, and fastened into the snowboard.

18. The snowboard training device as claimed in claim 16, wherein the predefined angle is an obtuse angle.

19. The snowboard training device as claimed in claim 16 comprising: the handle receiver further comprising a plurality of brace supports; each of the plurality of brace supports being perimetrically connected to the receiver body, extending downward to the top surface of the base plate; the plurality of brace supports comprising a first side brace, a second side brace, and an aft brace; and the aft brace being positioned along the aft end of the base plate.

20. The snowboard training device as claimed in claim 16 comprising: the handle grip comprising a pair of side handles and a center handle; the pair of side handles being configured for tilting the snowboard; and the center handle being configured for pushing and pulling the snowboard.